

Overview: Publishing & Deployment in ASP.NET Core MVC

Publishing and deployment are the final phases of the software lifecycle where your application moves from **development (local machine)** to a **production environment (server/cloud)**.

◆ What is Publishing?

Publishing is the process of:

- Compiling your ASP.NET Core MVC application
- Bundling all required files (DLLs, configs, static files)
- Preparing it for hosting

👉 Output: A **publish folder** containing optimized, deployable files.

◆ What is Deployment?

Deployment is the process of:

- Moving published files to a **server**
- Configuring hosting environment
- Making the app accessible via URL

👉 Example environments:

- IIS (Windows Server)
 - Azure App Service
 - Linux + Nginx
 - Docker containers
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ASP.NET Core MVC Deployment Architecture

Key Components:

- **Kestrel Server** → Internal web server
- **IIS / Nginx** → Reverse proxy
- **SQL Server** → Database

- **AppSettings.json** → Configuration (connection strings)
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✖ Step-by-Step: Publish ASP.NET Core MVC App

✔ Step 1: Prepare the Application

- Ensure:
 - No build errors
 - Proper connection string in appsettings.json
 - Environment-specific configs

```
"ConnectionStrings": {  
  "DefaultConnection": "Server=YOUR_SERVER;Database=YourDb;Trusted_Connection=True;"  
}
```

✔ Step 2: Configure Environment

Use:

```
"ASPNETCORE_ENVIRONMENT": "Production"
```

✔ Step 3: Publish using Visual Studio

Steps:

1. Right-click project → **Publish**
2. Choose target:
 - Folder
 - IIS
 - Azure
3. Select:
 - Deployment Mode (FDD/SCD)
4. Click **Publish**

👉 Output folder:

bin/Release/netX.X/publish/

✅ Step 4: Publish using .NET CLI (Alternative)

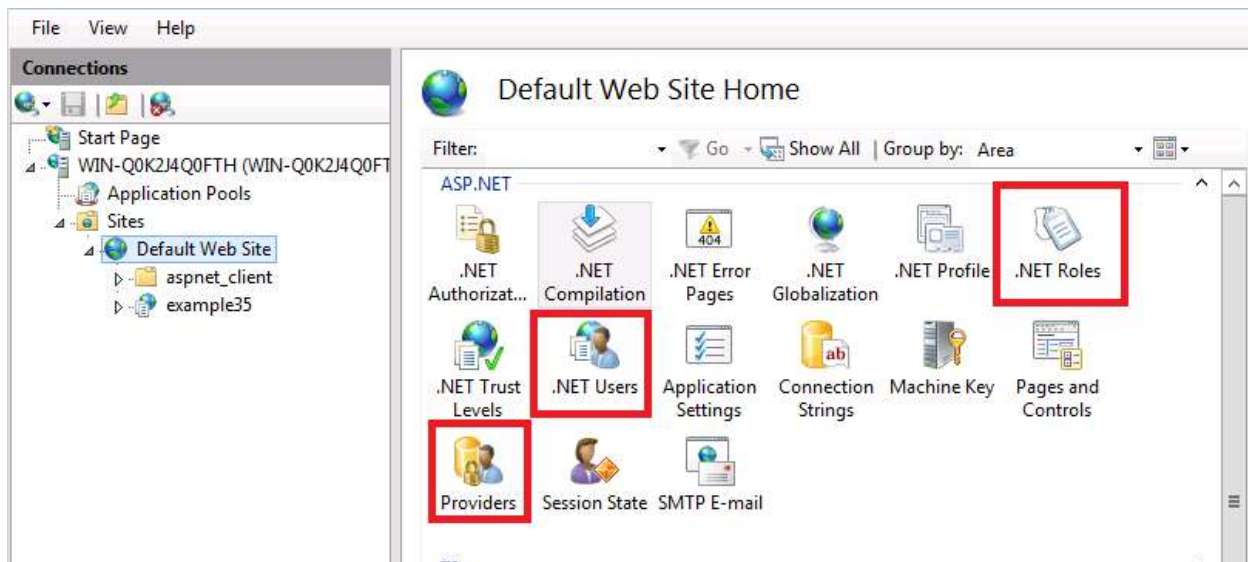
```
dotnet publish -c Release -o ./publish
```

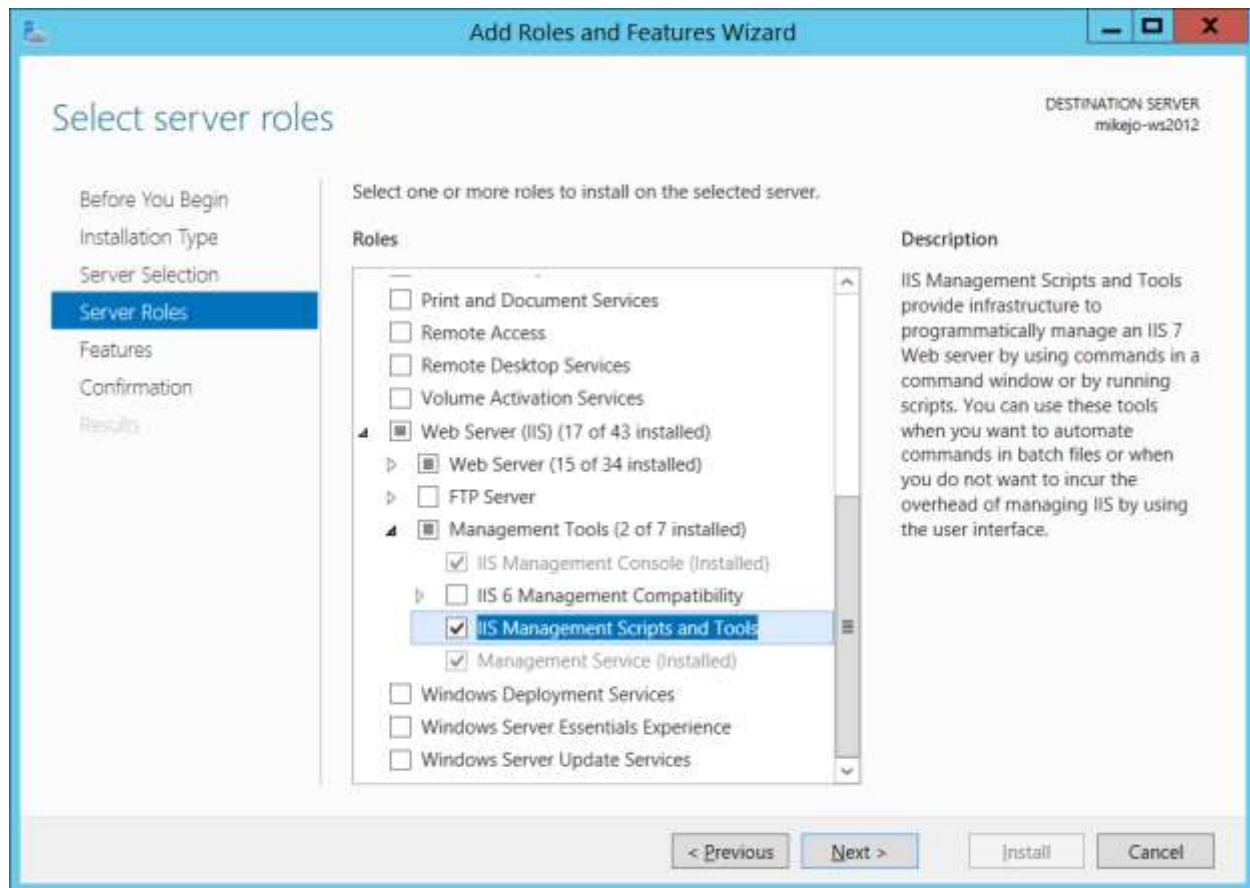
Step 5: Update Connection String

For production:

```
"DefaultConnection": "Server=SERVER_NAME;Database=MyAppDb;User  
Id=sa;Password=yourpassword;"
```

🔗 Step-by-Step: Deploy to IIS (Windows Server)





✓ Step 1: Install IIS

- Go to:
 - Control Panel → Programs → Turn Windows features on/off
 - Enable:
 - IIS
 - ASP.NET Features

✓ Step 2: Install Hosting Bundle

Install:

- .NET Hosting Bundle

👉 Required for running ASP.NET Core apps on IIS

✅ Step 3: Create IIS Website

1. Open **IIS Manager**
2. Right-click → **Add Website**
3. Configure:
 - Site Name
 - Physical Path → Published folder
 - Port (e.g., 8900)

✅ Step 6: Run Application

- Open browser:

http://localhost

☁ Optional: Deploy to Azure

Using Microsoft Azure:

Steps:

1. Create App Service
2. Configure SQL Database
3. Publish from Visual Studio → Azure
4. Update connection string in Azure portal

⚠ Common Deployment Issues

Issue	Cause	Solution
500.30 Error	Hosting bundle missing	Install .NET Hosting Bundle
DB Connection Failed	Wrong connection string	Verify credentials

Issue	Cause	Solution
Static files not loading	Missing wwwroot	Check publish output
Migration not applied	DB not updated	Run dotnet ef database update
